

Calculus I. Partial 1: Limits and the derivative as a limit process

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Jan-May 2026. Recent compilation: January 10, 2026

Class conditions

- There are no dumb questions
- Cellphones at the professor's desk at the beginning of class
- If you want to use the cellphone, take it from the desk and go outside
- Bring notebook
- All activities and homeworks are delivered by hand
- Late deliveries: Two days after due date with grade over 80
- No electronic devices
- Web assign is for home
- Most of homeworks are webassign and exam corrections

How to solve exam problems

Concepts

Practice

Feedback

Recap

- Notation for the course
- Types of functions (Algebra and graph forms)
- Evaluation of the function
- Function composition

Type of functions

- Linear
- Quadratics
- Cubic
- Exponential and Logarithmic functions
- Radical functions
- Root functions
- Two dimensional functions: Conic Sections

Algebra toolkit

Algebra manipulation

The big picture: Motivation

A nonsense that I'm building Sometimes I say that mathematics is a language. When we overthink how language can be describe, we can say that have words and those words can be classified as verbs, adjectives, nouns and adverbs. And also it have some grammatical rules. So, in math, the words and grammatical rules are provided by algebra, meanwhile the thing that help us to classified the mathematical objectives into groups is calculus.

Why see limits? Why bother to learn calculus?

Actually the motivation At some point in the history of humankind there was two open questions: How do you find the slope of a curve at a single, exact point? and How do you measure the area of a shape with a curved boundary? The first question was asked by the greeks. They were studying lines that only touches once a circle or most commonly known as the tangent line to a circle. However, for other kind of graphs it was a mystery, like for a straight line, a parabola or an ellipse. After some centuries, the answer to this question led us to the development of the derivative. On the other hand, the second question led to the creation of the integral. In a more specific tale, Archimedes was a master of computing areas under parabolas, however its method was a headache. Some years later, astronomers like Kepler need to calculate areas of planetary orbits and this was solved by the definition of an integral.

Integral allow us to give a quantitative answer to questions like "How much in total?", meanwhile, derivatives allow us to give a quantitative answer to questions that ask "How much change in total?".

And we can combine both, "How much change in total the total amount of something?"

Ideas to sketch the class Show them how to read the math. To process of demonstrations class. One to show the procedure to solve the problem. The other to show what is the graphical representation of the procedure. The second one is to read the math.

References

James Stewart, Dan Clegg, and Saleem Watson. *Calculus early transcendentals* James Stewart, Daniel Clegg, Saleem Watson. Cengage, 2024.